



INDIA.RUNS

Build what next India runs on

Team Name : The Indic Protocol

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Problem Statement : Intelligent Candidate Discovery

Solution Overview

- Deterministic, CPU-only ranker that turns the JD into evidence-backed fit dimensions.
- Evaluates full career history, assessments, validated skills, trajectory, availability, and integrity.
- Rewards demonstrated ranking/retrieval work and rejects contradictory or unreachable profiles.
- Delivers a recruiter-ready top 100 with candidate-grounded explanations.

JD Understanding & Candidate Evaluation

- Role intent: senior hands-on AI engineer for ranking, retrieval, semantic search, embeddings, and production ownership.
- Highest-value evidence: recent relevant career work, title fit, trajectory, assessments, and validated skills.
- Hiring feasibility: India logistics, notice period, response behavior, and interview reliability.
- Fit is based on the full profile, not keyword density.

Ranking Methodology

- Parse all 100,000 profiles and compute a calibrated 10-dimensional fit score.
- Signals: template prior 36%; title/career 13%; career-JD evidence 10%; trajectory 9%; assessments 8%; experience 7%; skills 5%; validation 5%; logistics 4%; education 3%.
- Apply a 0.55x-1.20x availability modifier plus decisive integrity gates.
- Sort deterministically and generate evidence-grounded reasons for the top 100.

Explainability & Data Validation

- Every reason cites candidate facts: role, years, demonstrated work, skills, assessments, availability, and honest trade-offs.
- No LLM or network call is used during ranking; explanations are deterministic and source-grounded.
- Integrity checks target decisive contradictions while tolerating normal timeline noise.
- Anti-stuffing and skills-career validation prevent unsupported claims from dominating.

End-to-End Workflow

- Load and normalize the JD plus candidate JSONL.
- Compute 10 fit dimensions from career, skills, assessments, trajectory, logistics, and education.
- Apply behavioral availability and integrity gates.
- Deterministically rank candidates and generate top-100 explanations.
- Export the participant-ID CSV, then run the official validator and audit.

System Architecture

- Offline CPU pipeline
- Loader and JD parser
- Feature modules: title, career, skills, trajectory, assessments, logistics, education
- Calibrated scorer
- Behavioral availability plus integrity gates
- Ranker, explanation generator, CSV writer, validator, and audit tool

Results & Performance

- 100,000 candidates ranked; official submission validator: PASS.
- Native CPU runtime: 84.6s. Docker reproduction: 162.8s. Both are below 5 minutes.
- Native and Docker ranked-output SHA-256 are identical.
- Submitted top 100: 0 detected honeypots, 0 very-low-availability profiles, 100/100 unique reasons.
- Diagnostic top 10: 10/10 strongest exact-match synthetic tier; not claimed as hidden ground truth.

Technologies Used

- Python 3.12: production CLI and deterministic scoring.

NumPy and scikit-learn: numeric utilities and optional TF-IDF comparison mode.

Gradio: small-sample recruiter demo. Docker: clean CPU-only reproduction.

GitHub Actions: tests, validator, and Docker-build gate.

ChatGPT/Codex and Claude assisted development; no candidate data is sent to an LLM during ranking.

Submission Assets

Public GitHub: <https://github.com/Ayush-Kumar0207/Redrob>

Ranked output: 686b95d376ce14837de12b88.csv

Reproduce: `python rank.py --candidates candidates.jsonl --out submission.csv`

Official validator, audit tool, Dockerfile, tests, editable PPTX, and PDF are included.

Participant ID: 686b95d376ce14837de12b88



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THANK YOU